



Plant Lipoxygenases and Their Role in Plant Physiology

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Abstract

Several enzymes play a key role in plant growth and development. Among these, lipoxygenases family of enzymes are important. Lipoxygenases are a ubiquitous family of non-heme iron enzymes widely distributed in plants, initiate hydroperoxidation of polyunsaturated fatty acids containing *cis*, *cis*-1,4-pentadiene moieties, and produce phytooxylipins. Oxylipins including green leaf volatiles and hormone jasmonic acid play a significant role in physiological processes, seed germination, fruit ripening, and senescence. Detailed studies have shown that LOX and its products accumulate transiently upon developmental or environmental stimuli. LOXs play a key role in defense responses against biotic and abiotic environmental stresses such as fungi, bacteria, virus, nematodes, birds, insects, and rodents. As a result, these enzymes are the natural topics of study when scientists investigate plant response to stresses. In this review, we have concisely focused on the analysis of new information concerning plant LOXs and their role on the physiology of plants.

Keywords Lipoxygenase · Oxylipins · Pathogen attack · Abiotic stress · Wounding · Senescence

Introduction

Lipoxygenases (LOXs; EC 1.13.11.12) are non-heme, non-sulfur oxidoreductases with iron or manganese as a cofactor that catalyze the oxidation of linoleic acid (LA, C18:2 ω 6), linolenic acid (ALA, C18:3 ω 3), and arachidonic acid (AA, C20:4 ω 6), thereby converting it to hydroperoxy octadecadienoic acid (HPOD), hydroperoxy octadecatrienoic acid (HPOT), and hydroperoxy eicosatetraenoic acid (HPOET), respectively (Brash 1999; Andreou et al. 2009). LOXs enzymes are widely distributed in mammals, plants, fishes, mosses, bacteria, yeast, fungi, corals, algae, and mushrooms, signifying that LOXs are crucial enzymes in biological systems (Joo and Oh 2012). Various signaling molecules called oxylipins are synthesized by LOXs' action, particularly, C6-volatile compounds and jasmonates are major signaling molecules in plants and play a significant role in their physiology.

LOXs present in the living organisms are classified according to their regiospecificity, size, and linkage of the enzyme with other proteins. According to the regiospecificity, the LOX superfamily is classified into 5-, 8-, 9-, 10-, 12-, 13-, 15-LOXs called classical LOXs. Among these classical LOXs, the major enzymes are 9- and 13-LOXs in plants, and 5-, 12-, and 15-LOXs in animals.

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LOX Pathway in Plants

The LOX pathway shows some differences in plants and animals. The key variance among them is the starting reaction of signaling molecule formation. According to the regiospecificity, plant LOXs are classified into two major subfamilies, 9-LOXs and 13-LOXs. In a few cases, LOXs with dual specificities such as 9/13-LOX or 13/9-LOX were also identified depending on the product principally formed (Hughes et al. 2001).

9-LOX Pathway

9-LOX is a non-heme, iron-containing enzyme with a single polypeptide chain having approximately 741–886 amino acids (Joo and Oh 2012). In the 9-LOX pathway, PUFAs are converted to major metabolites such as 9-HPOD, 9-HPOT, and 9-KOT (9S-keto octadecatrienoic acid) (Fig. 1a). Most of the products derived from this pathway are involved in the physiological functions of plants predominantly in defense against various pathogens. In contrast to the 13-LOX-derived signals, 9-LOX-derived signals initiate in several cases because of the less complex pathways and also the oxylipins produced from 9-LOX are structurally related to, but different from, the 13-LOX-derived oxylipins (Howe and Schilmiller 2002).

13-LOX Pathway

13-LOX is a non-heme, iron-containing single polypeptide composed of approximately 896–941 amino acids (Joo and Oh 2012). 13-LOX pathway catalyzes the conversion of PUFAs into 13-HPOT, which is supplementarily metabolized to plant signaling compounds such as jasmonates and green leaf volatiles (GLVs) (Fig. 1b). In plants, most of the LOXs reported till now belong to 13-LOXs, which play a pivotal role in the synthesis of jasmonates. Several downstream pathway branches utilize the products of 13-LOXs; however, at present, the best-characterized enzymes are members of the CYP74 family such as AOS, HPL, DES, and allene oxide cyclase (AOC) (Brash 2009), and also these enzymes showed close relationship with each other (Wasternack and Feussner 2018). Since the last decade, recognized signaling molecules other than jasmonates are continuously increasing, and the 13-LOX pathway is still the best-studied example.

Physiological Functions of Plant LOXs

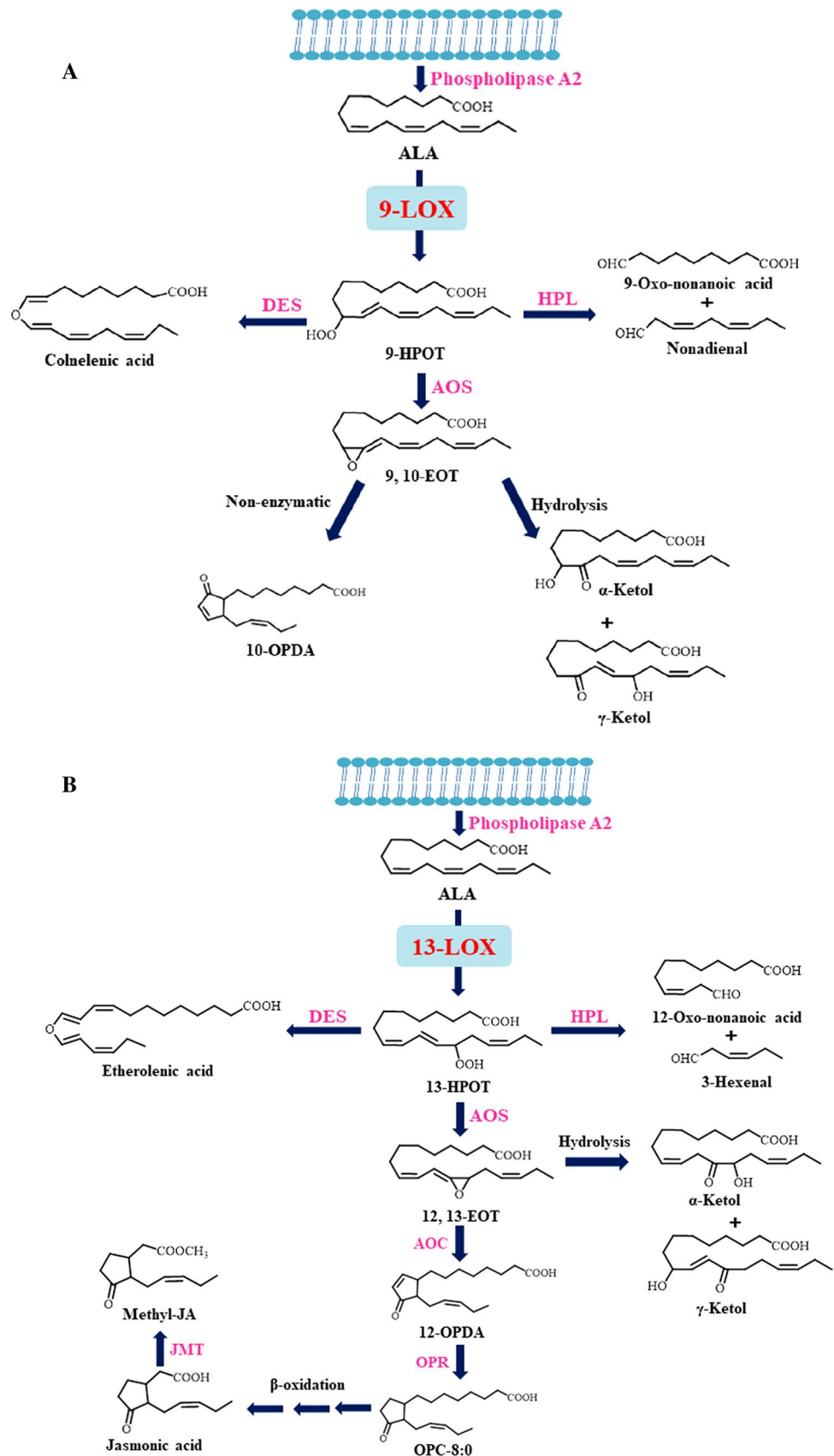
LOXs have been identified in various cellular parts that produce signaling molecules and these are involved in diverse functions (Fig. 2), which include growth and development, seed germination, response to biotic and abiotic stress and

wounding, fruit ripening, senescence, cell death and synthesis of JA and ABA.

Role of LOXs in Germination and Seedling Growth

LOXs are present in the seeds, act as vegetative storage proteins (VSPs) and play an important role during seed maturation and seedling growth (Siedow 1991). During germination, lipids are mobilized by a different set of LOXs. In the early stages of seedling growth, several new LOXs are induced, as reported in different plant species such as pea, watermelon, French bean, Lupin, finger millet, mung bean, carrot, lentil, barley, cucumber, *Arabidopsis*, wheat, *Papaver somniferum*, *Phoenix dactylifera*, rice, pepper, sunflower, and *Brassica napus*. However, in soybean during seed germination, lipids are degraded through an LOX-independent process, as the three LOX isoforms present in mature seeds disappear during the early stages of germination (Feussner and Wasternack 2002; Weber et al. 1999). However, proteome profiling analysis in rice and soybean germinating seedlings displayed that lipids were degraded through LOX-dependent pathway and the LOX could also help to eliminate the reactive oxygen species (ROS) during the rapid mobilization of reserves of soybean germinating seeds (Han et al. 2013). A major LOX L-4 appeared in soybean cotyledons after germination and produced both 13S-HPOD and 9(S)-hydroperoxy-10(E), 12(Z)-octadecadienoic acid from LA. Other than germination, *OsLOX2* in rice shows an effect on seed longevity during storage (Huang et al. 2014). Suitable repression of the *OsLOX2* gene may postpone the aging process throughout storage without conceding germination under normal conditions. Recently, the antisense suppression of LOX3 gene expression in rice endosperm revealed that the transgenic plants displayed a clear reduction in LOX transcript levels, and 9-HPOD from LA was expressed in low levels (Xu et al. 2015). The decrease in LOX specific activity in different bread wheat cultivars results in an effective decrease in grain lipid oxidation and extended grain storage period (Leenhardt et al. 2006). In *Vicia sativa*, three novel LOX isoenzymes appeared during germination, designated as P-1, P-2, and P-3. These three LOX isoenzymes show both dioxygenase and fatty acid lyase activities. Seeds of chickpea, lentil, broad bean and kidney bean contain two major LOXs: one synthesizes mainly 13-hydroperoxide from LA, whereas the other produces 9- and the 13-hydroperoxides and 9- and 13-ketodienes (Clemente et al. 2000; Sanz et al. 1993). Peanut seeds also contain three LOX isoenzymes with biochemical properties similar to the three soybean isoforms (Burow et al. 2000). Higher levels of 9-LOX expression were reported in *Zea mays* and almond during the seed developmental stage (Jensen et al. 1997; Santino et al. 2005). In cucumber, the physiological functions of LOXs within whole cotyledons were examined

Fig. 1 a The 9-LOX pathway in plants. The deoxygenation of ALA catalyzed by 9-LOX results in derivative 9-HPOT. It converted into other forms of DES, HPL, and AOS for an analogous set of reactions in the 9-LOX pathway and produced oxylipins. 9-HPOT 9S-hydroperoxy octadecatrienoic acid, 9,10-EOT 9S-epoxy octadecatrienoic acid, DES divinyl ether synthase, HPL hydroperoxide lyase, AOS allene oxide synthase, 10-OPDA 10-oxo-phytodienoic acid. **b** The 13-LOX pathway in plants. The ALA catalyzed by 13-LOX results in derivative 13-HPOT. It converted into other forms of DES, HPL, and AOS for an analogous set of reactions in the 13-LOX pathway and produced oxylipins, particularly plant hormone JA. 13-HPOT 13S-hydroperoxy octadecatrienoic acid, 12,13-EOT 13S-epoxy octadecatrienoic acid, 12-OPDA 12-oxo-phytodienoic acid, OPC-8:0 3-oxo-2-[2'-pentenyl] cyclopentane-1-octanoic acid, AOC allene oxide cyclase, OPR 12-OPDA reductase, JA jasmonic acid, JMT JA-carboxyl methyltransferase. The primary steps of JA biosynthesis (i.e., those catalyzed by LOX, AOS, and AOC) occur in the chloroplast, whereas the OPR and β -oxidation reactions take place in peroxisomes



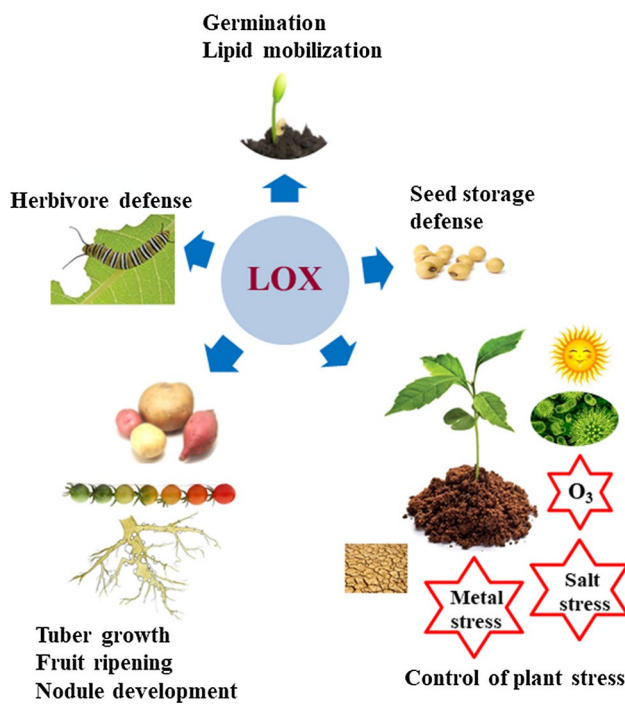


Fig. 2 Physiological functions of plant LOXs during plant growth and development

by determining the endogenous amounts of LOX-derived products and their oxylipins (Weichert et al. 2002). This was the first confirmation for the participation of a lipid-body LOX form in the production of volatile aldehydes. Similarities in the natural and accelerated aging-induced changes in the lipidome of *Arabidopsis* seeds were observed (Oenel et al. 2017). Identified multiple seed aging markers comprise many hydroxylated triacylglycerols, diacylglycerols, and phospholipids, and oxidized and non-oxidized fatty acids. These results suggested that LOX2 and LOX1 are responsible for the production of non-enzymatically and enzymatically oxidized free fatty acids, respectively.

LOXs are used as VSP in non-seed tissues, and are also present in flowers, pod walls, and germinating cotyledons. VSPs are involved in the developmental process and the accumulation of excess nitrogen products (Porta and Rocha-Sosa 2002). VSPs such as VlxB, VlxC, and VlxD are found in soybean leaves, para venial mesophyll (PVM) cells, and bundle sheath (BS) vacuoles using affinity-purified immunocytochemistry (Turner et al. 2011). In olive pollen grain germination, storage lipids are gradually processed during further steps of pollen tube growth. Functional characterization of the male gametophyte enzyme machinery during lipid mobilization revealed that the LOX enzyme plays a major role associated with phospholipase A, lipase (Greco et al. 2011). The role of the LOXs gene in tapetal cell membrane degradation of *Lathyrus undulatus* was identified during programmed cell death (Vardar and Meral 2011). In potato,

POTLOX-1 shows 9-LOX activity; it is expressed in tubers and roots and this was proved by transgenic experiments (Kolomiets et al. 2001). The relationship of patatins and LOX in stored tubers was studied and showed a negative correlation between LOX protein levels and tuber disease severity index and a positive correlation with tuber russet skin and LOX (Perla et al. 2014). The existences of LOX mRNA and proteins in legume nodules were identified in the developing stage, but their activity levels declined in nodules that reached their complete size (Porta et al. 1999).

Role of LOX in Pathogen Resistance

Plants respond to pathogens through the release of metabolites that have toxic or preventive effects, while others act as signaling molecules to induce defense responses. For biting–chewing herbivores, the LOX pathway has been proposed to play a key role in plant defense via major oxylipins such as 12-OPDA by the action of 13-LOXs, 10-OPDA, a positional isomer of 12-OPDA, and 10-oxo-11-phytoenoic acid (10-OPEA) by the action 9-LOXs (Crozier et al. 2000). Upon herbivory, LOXs lead to antiherbivore oxidative shift. This causes both direct and indirect oxidative injury to the herbivore as well as transduces the signals for provoking plant defense (Kaur et al. 2014).

The 9-LOX pathway plays a significant role in defense response to pathogen attack (Park et al. 2010), and its role was analyzed in *Arabidopsis* against *Pseudomonas syringae* and found that the 9-HOT induces cell wall modifications to limit pathogen invasion (Velloso et al. 2013). Moreover, the molecular events involved in 9-LOX signaling was analyzed using *nonresponding to oxylipins (noxy)*, a series of *Arabidopsis* mutants insensitive to 9-HOT, and found that cell wall modifications initiated by 9-HOT are under mitochondrial deteriorating control. In *Zea mays*, 10-OPEA along with 12- and 14-carbon cyclopente (a) nones, collectively called “death acids”, plays a significant role in the synthesis of a higher level of JA against fungal pathogen, *Cochliobolus heterostrophus* infection. It leads to direct phytoalexin activity, mediates defense gene expression and can promote cytotoxicity resulting in cell death (Christensen et al. 2015). *ZmLOX4* and *ZmLOX5*, the segmentally duplicated 9-LOX genes in *Zea mays*, were induced by fungal pathogens *Cochliobolus carbonum* and *F. verticillioides* and showed a unique resistance mechanism (Park et al. 2010). A 9-LOX mutant, *lox3-4* of maize, exhibited less root and mesocotyl necrosis caused by *Exserohilum pedicellatum* when compared with the wild-type *ZmLOX3* (Isakeit et al. 2007). The incompatible role of 9-LOX-derived oxylipins and ethylene pathway was determined in *Arabidopsis* via the 9-LOX derivative 9-HOT (López et al. 2011). These results established a role for the 9-LOX pathway in modulating oxidative stress, lipid peroxidation, and plant defense. A

rapid up-regulation of *CaLOX1*, a 9-LOX gene from pepper leaves, was detected during pathogen attack (Hwang and Hwang 2010). The activity levels of *CaLOX1* are more rapid in unsilenced pepper leaves than *CaLOX1*-silenced pepper leaves when infected with *Xanthomonas campestris* and *Colletotrichum coccodes*. Overexpression of *CaLOX1* in *Arabidopsis* showed greater resistance to *P. syringae*, *Hyaloperonospora arabidopsidis*, and *Alternaria brassicicola*. The role of *DkLOX3* was examined through overexpression in *Arabidopsis*: it exhibited improving resistance to *P. syringae* and *Botrytis cinerea* via adjusting ROS accumulation and cell death (Hou et al. 2018). LOX1, a 9-LOX gene present in *Arabidopsis* guard cells, plays a significant role in stomatal defense response compared with ABA (Montillet et al. 2013). It has been shown that oxylipins are potent inducers of stomatal closure. Mutations in LOX1 weaken stomatal closure and create tomato plants more vulnerable to *P. syringae* pv. Furthermore, strong 9-LOX activity has been found in the roots of *Arabidopsis* plants where 9-LOX products modify root construction (Velloso et al. 2007). In potato, 9-LOX oxylipins are involved in the defense process at the initial stage against *Phytophthora infestans* after 13-LOX products were identified in greater amounts in potato.

Increased levels of LOX activity have been reported in various plant–pathogen systems. Feeding of *Spodoptera exigua* larvae induces the expression of 9-LOX in *Zea mays* to a greater extent than 13-LOX, LOX3 upon inoculation of *Fusarium verticillioides* and *Aspergillus flavus* (Woldemariam et al. 2018). In rice leaves, LOX3 activity level was increased after infection with the blast fungus *Magnaporthe grisea*. In rice, Os9-LOX1 interacts with the 13-LOX facilitated a pathway to provide resistance against chewing and piercing-sucking herbivores. In almond-*Aspergillus carbonarius* interaction, 9-LOX produces 9-hydro (peroxy) fatty acids (HFAs), which are further metabolized into C9 aldehydes by the HPL branch pathway. Specific LOX activity levels were increased and correlated with the degree of host resistance in pearl millet during interaction with *Sclerospora graminicola* and in downy mildew-resistant plants. Peanut LOX is induced by *Aspergillus parasiticus*, *A. flavus* infections of mature cotyledons, and also numerous antimicrobial substances are produced via LOX reaction (Song et al. 2016).

Two plastidial 13-LOX genes, *ZmLOX6* and *ZmLOX10*, were induced in *Zea mays* by a fungal pathogen *C. carbonum* during compatible interactions (Gao et al. 2008b; Nemchenko et al. 2006). The responses of two *Arabidopsis* 13-LOXs, LOX3 and LOX4, to infection by *Meloidogyne javanica* and *Heterodera schachtii* were analyzed in transgenic *Arabidopsis* seedlings (Ozalvo et al. 2014). The results showed that LOX4 and LOX3 affect differentially with distinctive metabolic and signaling pathways and that

LOX4 plays a key role in governing plant defense against nematode infection. An increase in LOX2 specific activity in all leaves, developing seeds and pod walls in pigeon pea upon *Helicoverpa armigera* infestation (Kaur et al. 2014) and in groundnut upon infestation by *H. armigera* and *Aphis craccivora* was reported, but a substantial decrease in LOX activity with an increase in the development stages of cotton leaf due to *H. armigera* damage was detected. The effect of pea seed-borne mosaic virus (PSBMV) inoculation in *Pisum sativum* cultivar Raman and B99 increased LOX activity (Cerna et al. 2017). In tea plant, the *CsLOX1* is quick and extremely induced by the feeding of tea green leafhopper *Empoasca vitis*, whereas feeding by the tea aphid *Toxoptera aurantii* resulted in a pattern of alternating induction and suppression (Liu and Han 2010). The role of LOX in defense was determined in Brussels sprouts plants by the redox-active compound 1-phenyl-pyrazolidinone (phenidone), an inhibitor of LOX. It reduced the accumulation of signaling compounds and led to reduced herbivore-induced plant defense (Bruinsma et al. 2010). Papaya seedlings showed antifungal activity via 13-LOX produced hydroperoxides against *Phytophthora palmivora* by inhibiting the germination of sporangia and the growth of mycelia (Sujatha et al. 2012). Diverse LOX isoforms have been reported in potato during late blight, wounding, and pathogen infection. They are organ precise and are differentially controlled in expression during tuber growth. Overexpression of *TomLOXD* in tomato was found in JA biosynthesis during the attack of *H. armigera* and *B. cinerea* (Yan et al. 2013). *Suppressor of prosystemin-mediated responses8* (*spr8*) mutant plants are produced from a point mutation in the catalytic domain of *TomLOXD* displaying sequences of JA-dependent immune deficiencies and rigorously compromised resistance to *H. armigera* and *B. cinerea*. Also, silencing of the *TomLOXD* gene resulted in a clear reduction in the expression levels of defense genes *LeHSP90*, *LePR1* and *LePR6*. It reduced thermotolerance in transgenic tomato plants and increased its susceptibility to *Cladosporium fulvum* (Hu et al. 2015). LOX transcript levels were increased during pathogen attack in various plant species, such as in tomato by feeding of *Myzus euphorbiase*, in *Nicotiana attenuate* by *M. nicotianae* and *Manduca sexta*, and in cabbage by some caterpillars (*Pieris rapae*, *P. brassicae*, and *Mamestra brassicae*), spider mites (*Tetranychusurticae*), and locusts (*Schistocerca gregaria*), but not by aphids (*M. persicae*).

Hypersensitive response (HR) causes quick death of plant cells localized around the infection site. This limits further spreading of pathogen and plant cell damage. Therefore during this process, LOX products, mainly 9-LOX products, play a key role. The HR was examined in tobacco leaves through the production of oxylipin-reactive electrophile species (RES) adducts to glutathione (GSH) (Davoine et al. 2006). The oxylipin RES adducts to GSH are produced in

association with GSH consumption, increase in glutathione S-transferase activity, and the presence of cell death symptoms. In *Arabidopsis*, LOX1 displayed more antibacterial activity against *P. syringae* pv. tomato (Pst) infection; also pretreatment of LOX mutant plants with 9-LOX caused the production of 9-KOT and protected the plant tissues against bacterial infection (Vicente et al. 2012). The long-lasting, wide-ranging spectrum resistance, called induced systemic resistance (ISR) of biological control by plant growth-promoting rhizobacteria (PGPR) in tomato plants, revealed that the enhanced levels of *TomLOXD* and *TomLOXF* were noticed in *Pseudomonas putida* BTP1-inoculated plants before and 2 days after infection by *B. cinerea* (Mariutto et al. 2011). Principally, *TomLOXF* consumes more ALA and produces free antifungal oxylipins such as 13-HPOET and 13-hydroxy-octadecatrienoic acids (13-HOTrE). Increased levels of LOX activity in ISR was also identified in groundnut, *P. vulgaris*.

Effect of LOXs on Abiotic Stresses and Plant Hormones

Comparatively few studies have been carried out in plants to establish the role of LOX during abiotic stress. The *CaLOX1* plays a crucial role in osmotic, high salinity and drought stress responses through decreased accumulation of H_2O_2 , lipid peroxidation and high level expression of *RD20*, *RD29A*, *RD29B*, *P5CS*, *COR15A*, and *DREB2A* genes (Lim et al. 2015). *DkLOX3* shows more resistance to osmotic, high salinity and drought stress with an increase in germination rate, root growth, and up-regulation of stress-responsive gene expression, including *RD22*, *RD29A*, *RD29B*, and *NCED3*, and reduction in accumulation of H_2O_2 and superoxide radicals (Hou et al. 2015). The total LOX activity levels were significantly increased in finger millet under drought and nickel stress, influencing the antioxidative enzymes and alleviating these stress effects (Kotapati et al. 2017).

Studies were conducted to determine the involvement of LOX in lipid peroxidation under heavy metal stress in *Arabidopsis* (Skorzynska-Polit and Krupa 2003). Up-regulation of LOX activity was observed in *P. vulgaris* seedlings under Cd^{2+} stress, in sunflower leaf segments treated with Cu^{2+} . In olive fruit mesocarp, expression levels of 9-LOX, *Oep2LOX1* gene was regulated by temperature, light and wounding (Padilla et al. 2012), whereas 13-LOX expression was increased by low temperature and wounding (Padilla et al. 2014). A new chloroplast-targeted LOX in *P. vulgaris*, *PvLOX6* was extremely induced after wounding, non-host pathogen infection, and by signaling molecules such as H_2O_2 , SA, ethylene, and MJ (Porta et al. 2008). Dark chilling (4 °C) and successive photo-activation modulate expression and induce reversible association of *PcLOXA*, a member of

type 2 plastid-localized 13-LOX identified in *P. coccineus* (Mazur et al. 2018). In spite of the wounding effect, chilling stress obviously induced triggering of *PcLOXA* gene inside chloroplasts and interactions with the grana margins of thylakoids, signifying a distinct role of this gene in low-temperature stress response. The expression profiling of 12 LOX genes was analyzed in *Cucumis sativus* during fruit development, hormonal treatments, and abiotic stress (Yang et al. 2012). Particularly, *CsLOX9* is involved in lateral root growth under cold stress, and *CsLOX10* responds only to NaCl treatment. The effect of abiotic stresses and signal molecules (ABA, MJ, and SA) on the expressions of five LOXs (*CmLOX8*, *CmLOX10*, *CmLOX12*, *CmLOX13*, and *CmLOX18*) was analyzed in sweet melon leaves (Liu et al. 2016). *CmLOX10* and *CmLOX18* might participate in the biosynthesis of JA. *CmLOX12* is required in several responses, because it was induced by all the treatments except cold. *CmLOX10*, *CmLOX12*, and *CmLOX13* might participate in GLVs biosynthesis and plant defense response to stress (Liu et al. 2016). The induced expression of *OsLOX2* in *O. sativa* under drought stress (Du et al. 2013), *LOX1* in *Arabidopsis* under cold stress, a rapid increase in *CtLOX3* concentration after 4 h of salt stress in citrus plants (Arbona et al. 2010) was reported. Overexpression of *Capsicum annuum* L. LOX1 (*CaLOX1*) in *Arabidopsis* enhances drought tolerance by up-regulation of *DREB2A*, *RD20*, *RD29A* and *RD29B*, and salt stress by ABA and ROS production (Lim et al. 2015). Overexpression of *DkLOX1*, *DkLOX3*, and *DkLOX4* in *Arabidopsis* shows a positive role in tolerance toward abiotic and osmotic stress and also promoting the ripening and senescence through the lipid peroxidation and stress regulation genes' expression via regulating ROS accumulation (Hou et al. 2015).

LOX is also expressed in related plant hormones including JA, SA, ABA, and nitric oxide (NO). In *Caragana jubata*, up- and down-regulation of LOX was observed during the exposure of plants to abscisic acid (ABA), methyl jasmonate (MJ) and salicylic acid (SA) (Bhardwaj et al. 2011). A higher expression of LOX1 was induced by ABA and MJ in mature *Arabidopsis* plants. However, a high level of LOX2 in leaves and inflorescence, whereas low levels of expression in seeds, roots, and stems were observed, but LOX6 showed more expression in roots. In *Zea mays* two 9-LOX genes, LOX4 and LOX5, were induced by JA and SA, and two 13-LOX genes LOX10 and LOX11 were preferentially expressed, particularly LOX10 in response to wounding, JA, SA, ABA and cold stress, whereas LOX11 was induced only by ABA.

In the intricate hormone cross talks, SA targets LOX2 to suppress the JA biosynthetic enzymes. Schaffer et al. (2007) described LOX1, LOX3, and LOX7 gene transcription to be ethylene induced in the cultivar 'Royal Gala', while NO induces the expression of a LOX3 gene which leads to JA

production (Palmieri et al. 2008). Lemeza et al. (2010) studied the effects of different phytohormones on the LOX activity and amount of 9-LOX gene transcripts in potato. All phytohormones—gibberellin, auxin, kinetin, epibrassinolide, and SA suppressed the activity of LOX. In vegetation experiments, there were huge differences in the correlation value of LOX activity and 9-LOX transcripts level for each phytohormone. Out of all phytohormones, only auxin and gibberellin increased the 9-LOX gene transcript amount. Higher expression of LOX precedes JA accumulation upon elicitation (Bae et al. 2016; Rahimi et al. 2014, 2016). Upon chitosan treatment, LOX1-derived products accumulate in *Adelostemma gracillimum* seedlings (Li et al. 2012a), upon MJ treatment, transcript levels of LOX3, LOX4, and LOX5 in *Panax ginseng* (Bae et al. 2016), *VrLOX* in green gram (Kotapati et al. 2015), *EcLOX* in finger millet (Kotapati et al. 2016), *CsLOX1*, 2, 4, 8, 9 and 10 in cucumber (Yang et al. 2012), *CsLOX1*, 3, 7, and 10 in tea plant (Zhu et al. 2018), *DkLOX3* in Persimmon (Meng et al. 2017), *GhLOX1* in *G. hybridus* were up-regulated, but levels of *GhLOX1* under salicylhydroxamic acid (SHAM), *CsLOX1*, 3, 7, 10 *EcLOX* and *DkLOX3* under SA treatment were decreased (Lian et al. 2011; Liu and Han 2010). *ZmLOX6* in maize was induced by JA, but suppressed by ABA, SA, and ethylene (Gao et al. 2008b). *TcLOX1* and *TcLOX2* in *T. chinensis* were up-regulated by JA and ABA, but *TcLOX1* and *TcLOX2* were not influenced by SA (Li et al. 2012b).

Role of LOX in Wound Stress

The LOX-derived hydroperoxides and oxylipins have been involved in the local and systemic response to wounding stress and it was clarified using the expression of heterologously wound-responsive LOX cDNA sequence in *E. coli* and also by the observation of LOXs expression in various plant species.

Various free radicals such as ROS, lipid-derived carbonyl, and malondialdehyde through LOX pathway released in wounding could increase membrane damage, leading to localized cell death usually related to “hypersensitive reaction (HR)”. A significant role of LOX in the wounding-induced oxidative burst is defined in *Pisum sativum* seedlings. It was found that extracellular LOX may play a key role in the oxidation of fatty acids and also generate reactive electrophiles that direct signaling responses (Roach et al. 2015). During wounding, singlet oxygen is formed ($^1\text{O}_2$) by LOX reaction involved in the oxidation of either lipids or proteins. This mechanism of $^1\text{O}_2$ generation at the wounding site is proved in *Arabidopsis* leaves by catechol, an LOX inhibitor, resulting in $^1\text{O}_2$ formation and generation being exclusively limited to chloroplasts. LOX present in chloroplast is principally responsible for this $^1\text{O}_2$ formation (Prasad et al. 2017).

n-Hexanal and 12-oxo-trans-10-trans-10-dodecenoic acid, also known as traumatin or wound hormone, involve in plants’ response to wounding, and traumatic acid was found to improve cell proliferation at the wound site, suggesting that these act as signals for wound healing. The C6-aldehydes such as (Z)-3-hexenal, (E)-2-hexenal, (E)-2-hexenol and *n*-hexanal found in most terrestrial plants are involved in the plant defense process under wound stress condition (Matsui 2006). These C6-aldehydes are generally present in little amounts; however, in wounding or herbicide damage conditions, LOX causes rapid formation of these compounds in plants and controls plant defense by triggering wound- and herbivore-induced defense responses. In tea plant, *CsLOX2*, 3, and 4 mediate the production of C6-aldehydes and alcohols; *CsLOX6* and *CsLOX7* are responsible for JA synthesis (Zhu et al. 2018). Four LOX genes, *PcLOXA*, *B*, *C*, *D*, have been identified in *P. coccineus*, among these *PcLOXA*, *B*, and *D* were induced by wounding and participate in the accumulation of JA synthesis (Mazur et al. 2018). *OsLOX1*, a dual-specific LOX from rice rapidly increased and produced more amount of JA, (Z)-3-hexenal and colneleic acid during wounding and brown planthopper attack (Wang et al. 2008).

Four 13-type of LOXs such as LOX2, LOX3, LOX4 and LOX6 from *Arabidopsis* were analyzed, among them LOX2 and LOX6 were identified for a bulk amount of JA production in leaf and xylem during wounding (Chauvin et al. 2013). LOX3 mostly in circumfascicular parenchyma, while LOX4 in phloem-associated cells is expressed, but compared with LOX6 and LOX2, where LOX3 and LOX4 reduced JA production in the wounded leaf. In tomato, a 13-LOX enzyme TomloxC is involved in the synthesis of C5-aldehydes such as 1-penten-3-ol (pentenol) and 1-penten-3-one (pentenone) from 13-HPOT (Shen et al. 2014). The C5-aldehyde levels were increased in tomato leaves subsequent to mechanical wounding.

Podolyan and co-workers (Podolyan et al. 2010) observed a clear and differential change in LOXs gene expression upon wounding. *ZmLOX10*, *ZmLOX8*, a key 13-LOX enzymes (Christensen et al. 2013; Nemchenko et al. 2006), and *ZmLOX5*, a 9-LOX gene from corn (Park et al. 2010), were induced in defense to herbivory and wounding. A LOX from cucumber fruits (Zhao et al. 2010), *LOX-H1* and *LOX-H3* from potato leaves, *TomLOXD*, from tomato leaves (Yan et al. 2013), *PvLOX6*, from *P. vulgaris* (Porta et al. 2008), *PgLOX4* and *PgLOX5* from *Panax ginseng* (Bae et al. 2016) contribute to the wounding through the synthesis of GLVs or JA.

Role of LOX in Senescence and Fruit Ripening

Numerous studies have specified that a rise in LOX activity is a common feature of senescing plant tissues. During

senescence, LOX activity levels were raised and causes the production of superoxide radicals. It leads to an increase in lipid peroxidation levels. These studies proposed that LOX is the key regulator of lipid peroxidation and may contribute to membrane damage during senescence. A sharp up-regulation of LOX activity in cytosol as well as microsomes of senescing bean cotyledons, in *Iris versicolor* flowers just before senescence (Ahmad and Tahir 2018), a 13-LOX in *Alstroemeria peruviana* floral tissues (Leverentz et al. 2002), and both young and adult leaves of *Pisum sativum* during foliar spraying of 2,4-dichlorophenoxyacetic acid (2,4-D) causes an increased accumulation in lipid hydroperoxides. It leads to membrane semi-permeability loss and electrolyte leakage leading to senescence.

The massive damage to the chloroplast occurs during senescence; however, how the chloroplast breakdown occurs has not been properly answered earlier. Recently, a mechanism was proposed for the well-ordered destruction of chloroplast through natural and MeJA-triggered senescence in barley that involves a 13-LOX (Springer et al. 2016). According to Springer, Armin, et al. findings, at the time of senescence, 13-LOX stored in chloroplasts of senescing plants and triggered local membrane destruction and leakage of stromal constituents (Fig. 3). Interestingly, these findings are comparable to the action of mammalian 15-LOX during membrane destruction of reticulocytes and central fiber lens cells. The specific expression pattern and the exact role of persimmon *DkLOX3* during senescence was investigated via transgenic *Arabidopsis* leaves (Hou et al. 2015). At the

time of senescence, the overexpression levels of *DkLOX3* in *Arabidopsis* leaves were noticed and it causes more yellow color with loss of chlorophylls, expressively greater levels of membrane permeability for electrolyte leakage, a higher increase of MDA content, superoxide radicals including O_2^- , H_2O_2 and dead cells related with wild-type plants. The comparative proteomic analysis in *Zea mays* shows that LOXs levels were increased at initial leaf senescence (Wu et al. 2018). ZmLOX3, a maize 9-LOX knockout mutant, showed advanced senescence and reduced root length and plant height (Gao et al. 2008a). Changes of LOX activity levels in *Avena sativa* during both natural and dark-induced senescence (Yi et al. 2005), a 13-LOX in barley during JA production (Sharma et al. 2006), *SeLOX-5* in chayote during wilting and senescence (Del Ángel-Coronel et al. 2018), LOX3 and LOX4 genes in *Brassica napus* in early senescence (Ling et al. 2018), *TomLoxB* in tomato during the latest phase of ripening and senescence, in tea plant CsLOX1, a dual positional specific LOX (Liu and Han 2010), and RLOX1 from the rose petals were greatly expressed during senescence with increased level of superoxide radicals. Delaying of senescence is observed in *Paris polyphylla* due to the retarding effect of gibberellic acid (GA3). It triggered an improvement of internal GAs (GA4 + GA7) levels, resulting in the delay of shoot senescence, which was endorsed as principally due to the suppression of LOX activity. Usage of melatonin in peach fruit postponed senescence and controlled ROS metabolism via decreasing the LOX activity (Gao et al. 2016). In tomato, postharvest life was

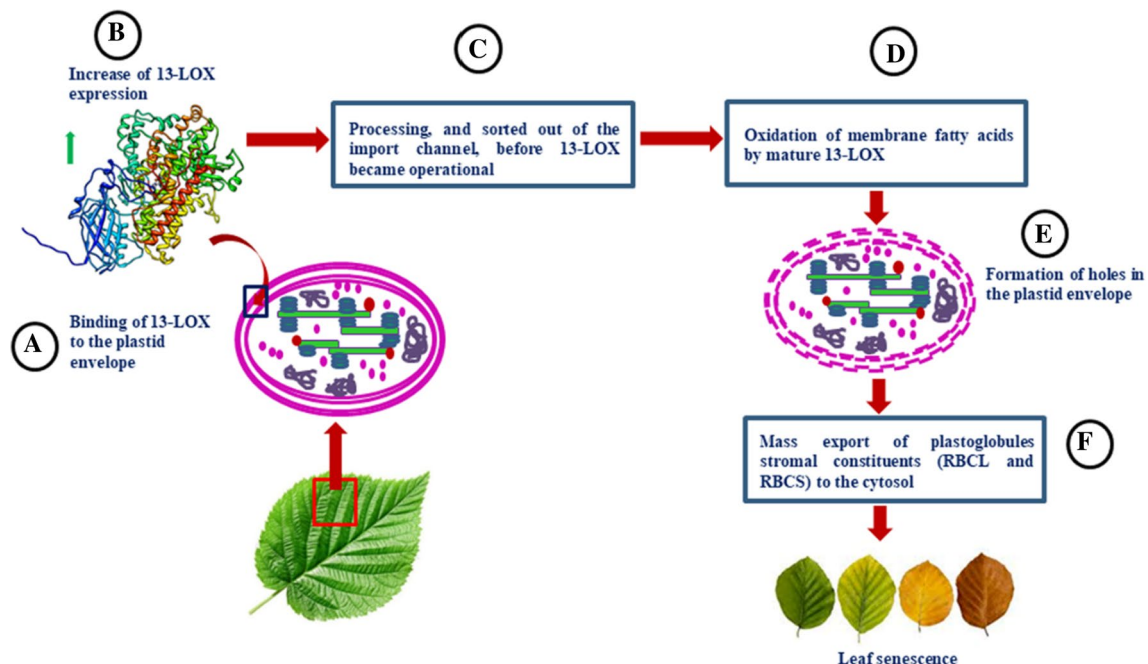


Fig. 3 Senescing process mediated by LOX action

increased under the silencing of *TomloxB* and delayed in fruit senescence (León-García et al. 2017). Decreased level of *BoLOX1* from broccoli heads was noticed when treated with cytokinins, modified atmosphere, hot air, UV-C or white light. It showed a delay in chlorophyll degradation and delayed senescence in broccoli during postharvest treatment (Gomez-Lobato et al. 2012).

An endogenous JA even plays a role in senescence and hence it is called a senescence enhancer. One higher level of JA and related oxylipin levels in senescing leaves were detected compared to non-senescing ones during natural senescence and upon induction of senescence-like phenotypes by dark treatment and flotation on sorbitol (Seltmann et al. 2010). The active role of JA in senescence was detected in *Arabidopsis* (He et al. 2002), and various genes were differentially activated to catalyze the reactions of the JA biosynthetic pathway during leaf senescence in *Arabidopsis*; among these genes cytoplasmic LOX-1 was strongly up-regulated (Schommer et al. 2008).

Apart from senescence, LOX plays a significant role in fruit ripening for controlling quality and taste in several climacterics and non-climacteric fruits (Del Ángel-Coronel et al. 2018; Schiller et al. 2015). Del Ángel-Coronel et al. (2018) confirmed up-regulation of LOX activity in association with fruit quality parameters and identified five LOX isoforms, *SeLOX-1*, *SeLOX-2*, *SeLOX-3*, *SeLOX-4*, and *SeLOX-5*, in *Sechium edule*. They confirmed that the rise of *SeLOX-2*, -3, and -4 in fruit coincided with the critical water loss, reduced LA, ALA, and soluble solid contents, and increase in fruit firmness. Based on these observations, they suggested that the activity of *SeLOX-2*, -3, and -4 could be associated with fruit ripening. Highly distinct expression patterns of *MdLOX1* and *MdLOX5* genes from apple (Vogt et al. 2013), *CmLOX18* from melon (Zhang et al. 2014), LOX from *Solanum quitoense* (Iguarán et al. 2018), and *MiLOX* from *Mangifera indica* (Deshpande et al. 2017) are responsible for the production of fruit aroma volatiles and late development during fruit ripening. There were increased levels of LOX specific activity in olive during fruit development (Palmieri-Thiers et al. 2009). 12 differentially expressed LOXs from cucumber have diverse patterns of expression in the exocarp, endocarp, and pulp at day 5 after anthesis in relation to the different stages during fruit ripening (Yang et al. 2012). Four LOXs from peach such as *PpLOX1*, *PpLOX2*, *PpLOX3*, and *PpLOX4* levels were increased during fruit ripening. Of these LOXs, *PpLOX2* and *PpLOX3* had comparatively high transcript levels firstly, followed by a decline with fruit ripening, whereas the levels of *PpLOX1* and *PpLOX4* transcripts were up-regulated by accumulated ethylene production (Han et al. 2011; Zhang et al. 2010). *SmLOXD*, *SmLOXB*, and *SmLOX5* levels were increased in *Solanum muricatum* Aiton, and C9 volatiles such as (E)-2-nonenal and (E, Z)-2,6-nonadienal were

produced during immature stages with a related decrease when the fruit ripened, whereas C5 and C6 volatiles were formed throughout ripening (Contreras et al. 2017). In tomato, postharvest life was increased in fruit ripening under the silencing of *TomloxB* (León-García et al. 2017). Reduced LOX activity levels were detected in the delayed ripening and senescence of apple pulp by the exogenous apply of chlorogenic acid (Xi et al. 2016).

Concluding Remarks and Future Perspectives

The LOX family is diverse and isozymes often differ in terms of activity, substrate specificity, and product formation, even in the same organism. LOXs and their oxidation products of PUFAs play a major role in metabolism in plants and have an effect on their growth and development and tolerance to biotic and abiotic stresses. Our review provides recent information for the plant LOXs along with earlier findings, but due to space constraints, we have not covered some of the information related to plant LOXs.

Over the past few years, a number of authors have succeeded in cloning, purification, and characterization of LOX enzymes to gain better understanding. Also needs to study their role in synthesizing green leaf volatiles, an associated mechanism with other plant hormones during plant growth and development. Particularly, in oxylipins synthesized by plants, LOXs need more attention due to their significant role in plant physiology. Identification of new alleles active in oxylipin formation will shed new light on the diversity among oxylipins.

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